

AI Governance in Hospital Management

The NHS Patient Flow & Bed Management Case

Pradyumna Reddy Dereddi · Nishanth Reddy Dereddi · Sireesha Rodda

GITAM Deemed University, Visakhapatnam, India

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Agenda

01

Background & Problem Statement

Why AI governance in NHS hospital management matters

02

The NHS Case Study

UHB, Leeds Teaching Hospitals & PFIT

03

Governance Analysis

Regulation, transparency, equity, accountability

04

AI Techniques

Predictive modelling, NLP, explainability, optimisation

05

Proposed Framework

Four-pillar governance model

06

Conclusions

Key takeaways & future directions

Why AI Governance in Hospital Management?

The Challenge

- Hospitals face real-time complexity across clinical and operational layers
- Chronic inefficiencies in bed management cascade into delayed care & rising costs
- NHS serves 67 million+ people with 1.4 million staff under resource constraints
- AI offers predictive tools — but deployment is not a neutral technical act

67M+

NHS patients served

1.4M

NHS staff in England

0.5 days

Avg LoS reduction (UHB)

100K+

Annual admissions at UHB

NHS: AI for Patient Flow & Bed Management

University Hospitals Birmingham (UHB)

AI platform on Microsoft Azure + Epic EHR generates real-time discharge likelihood scores. Dashboards enable 24–48hr advance planning of patient movements, shifting teams from reactive to predictive management.

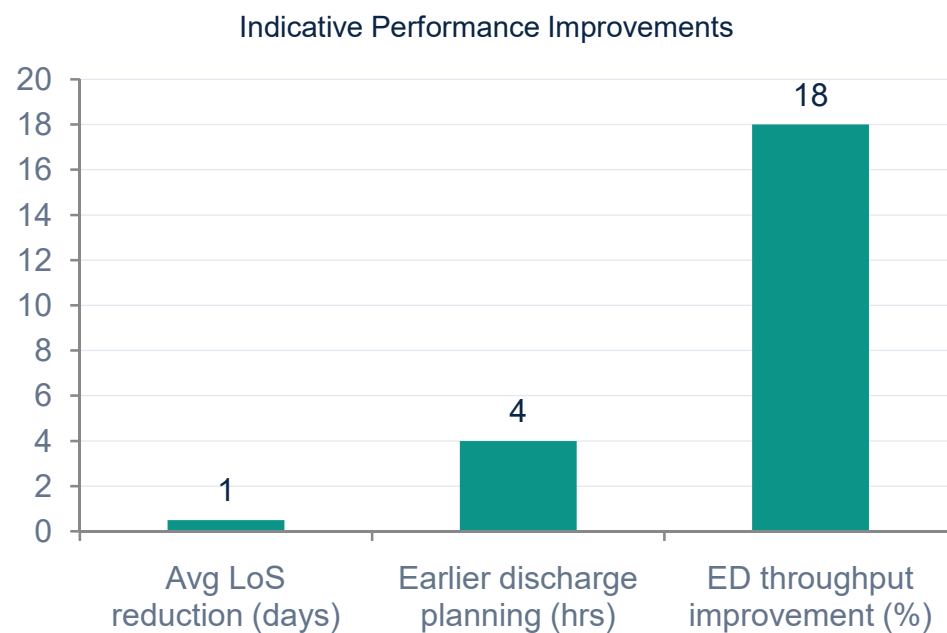
Leeds Teaching Hospitals

AI bed management system improved emergency department throughput by increasing upstream bed availability. Predictive alerts allowed ward teams to initiate discharge planning several hours earlier than baseline.

NHS Patient Flow Intelligence Tool (PFIT)

Extends predictive analytics to regional and national level — enabling demand modelling across entire health economies for system-level commissioners.

Reported Outcomes & Evidence Gaps



⚠ Evidence Gaps

Most evaluations are internal, single-trust, and non-randomised. Gains vary significantly by data quality and staff adoption (Nuffield Trust, 2022).

Context Dependency

Improvements are NOT automatic. Success depends on: data quality, organisational culture, and structured staff engagement from the outset.

Four Dimensions of AI Governance

01

Regulatory Framework

MHRA · NHS Digital · CQC govern AI tools across fragmented mandates. The joint AI & Digital Regulations Service is consolidating guidance but unified pathways are still needed.

02

Transparency & Explainability

GBM and neural network models are opaque. SHAP values surface feature contributions in UHB dashboards, but implementation remains inconsistent across trusts.

03

Equity & Bias

BAME patients face historically differential care. No mandatory equity audits exist for operational AI. Bias can be encoded and amplified without disaggregated performance reporting.

04

Accountability & Staff Engagement

Clinicians retain decision authority. UHB's co-design model built trust — proving staff engagement is not a communication afterthought but a governance requirement.

AI Techniques in Patient Flow & Bed Management

Predictive Modelling

- XGBoost / LightGBM (GBM)
- Logistic Regression (baseline)
- Cox Proportional Hazards (time-to-discharge)

Deep Learning & NLP

- LSTM / RNN (time-series vitals)
- Transformer models (clinical notes)
- NER & text classification

Explainability

- SHAP — feature attribution
- LIME — local approximation
- Both model-agnostic

Demand Forecasting

- ARIMA / Prophet (time series)
- Ensemble demand models
- Reinforcement Learning (emerging)

Optimisation

- Linear / Integer Programming
- Bed allocation constraints
- RL-based dynamic policies

Data Infrastructure

- Federated Learning (privacy)
- Transfer Learning (small trusts)
- Cross-trust gradient sharing

Four-Pillar Governance Framework



↻ Living Governance — continuous monitoring feeds back across all four pillars

Governance System Architecture

Tier 1 — Regulatory oversight

MHRA · NHS Digital · CQC · AI Reg. Service



Tier 2 — Trust-level governance

AI Committee · Algorithmic Audit · Procurement Standards



Tier 3 — AI System layer

Predictive Model · Explainability (SHAP) · Operational Dashboard



Tier 4 — Clinical & operational

Clinicians · Bed Managers · Staff Feedback Loops

Living
governan
ce
feedback

Staff Feedback Mechanisms

Frontline staff are the most important source of AI performance intelligence.

01

In-system flagging

"Flag this prediction" button in dashboards — creates audit trail & feeds model monitoring.

02

MDT review forums

Scheduled sessions where ward staff raise pattern anomalies — feed into governance committee.

04

Co-design cycles

Ongoing staff involvement at each update cycle — not just at launch. UHB model proved this works.

05

Anonymous reporting

Speaking-up guardian channel — removes fear of challenging institutional AI investments.

CONCLUSIONS

Key Takeaways



AI delivers measurable gains

UHB: ~0.5 days LoS reduction. Leeds: improved ED throughput. Gains are real but context-dependent.



Governance is the decisive factor

Transparency, equity auditing, accountability, and staff engagement determine whether AI serves the public interest.



Critical gaps remain

No mandatory equity audits for operational AI. Fragmented regulation. Proprietary black-boxes resist scrutiny.



Proposed framework

Proportionate regulation · Equity auditing · Algorithmic transparency · Living governance.



Future directions

Independent multi-site RCTs. Mandatory vendor transparency standards. Reinforcement Learning pilot evaluation.

Thank You

Questions & Discussion

Corresponding author: srodha@gitam.edu

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"Effective AI governance is not a compliance exercise — it is a continuous institutional commitment to safety, equity, and patient welfare."